

MPLADS IntelliTrack System

Comprehensive Service Architecture Aligned with the Complete Government Lifecycle

Domain: GovTech & Public Infrastructure

Architecture: 6 Pillars & 22 Core Services

Core Engine: AI-Powered Governance

THE COMPLETE SERVICE LIFECYCLE CHAIN

01. Recommendation → 02. Sanction → 03. Execution → 04. Evidence → 05. Financial Monitoring → 06. Anomaly Detection → 07. Risk Assessment → 08. Human Action

Note: AI acts as an intelligent supporting and analytical layer within a rigorous administrative workflow, ensuring transparent decision-making without black-box automation.

Architecture Grouped by 6 Major Pillars

01 — Project Management & Foundation

Pillar 1

1. 🏛️ Digital Project & Recommendation Management

Manages the complete digital record of an MPLADS work from inception.

- MP work recommendation & project details/specifications
- Estimated cost, location, beneficiary information & sanction details
- Implementing Agency and contractor mapping with Unique Project ID

2. 👥 Role-Based Access Control

Provides tailored access according to user role and administrative jurisdiction.

- **MoSPI:** National level overview
- **State & District Authorities:** State/District level monitoring
- **MP / MP Office:** Constituency-level tracking
- **Implementing Agency & Contractor:** Assigned works and contracts only

02 — Digital Evidence & Field Capture

Pillar 2

3. 📍 Geo-Tagged Project Verification

Validates whether submitted project evidence corresponds strictly to the approved project location.

- In-app camera capture with GPS coordinates and timestamping
- Project geofence and location-distance verification with mismatch alerts

4. 📷 Digital Progress Evidence

Creates a structured evidence trail for project execution across stages.

- Mandatory progress photographs from multiple angles and progress-stage submissions
- Timestamped evidence history, previous-vs-current comparison & duplicate/reused image detection

20. 🗺️ GIS-Based Project Visualization

Displays projects geographically with risk-based color-coding (Green: Normal, Yellow: Watch, Orange: Review, Red: High Risk).

03 — Financial Intelligence & Benchmarking

Pillar 3

6. 📄 Intelligent Document & Invoice Processing

Uses OCR and Document AI to automatically extract information from uploaded bills and invoices, reducing manual data entry.

Invoice → Vendor → Item → Quantity → Unit Rate → GST → Total

7. 💰 Smart Cost Benchmarking

Compares proposed or approved project costs with comparable historical projects factoring in location, materials, labour, and technical specs.

Expected range: ₹42-46 lakh | Proposed: ₹50 lakh | **Deviation: +12.4%**

8. 💳 Financial & Expenditure Monitoring

Continuously monitors sanctioned amounts, estimated costs, expenditure, payments, remaining funds, and expenditure trends to identify unusual financial behaviour.

9. 📄 Bill & Material Rate Anomaly Detection

Analyzes individual bills and material costs against standard benchmarks.

Reference rate: ₹20-23/unit | Submitted rate: ₹30/unit → ● **Material Cost Anomaly – Verification Required**

04 — AI Risk Intelligence & Anomaly Detection

Pillar 4

5. 🤖 AI-Based Image & Progress Verification

Computer vision supporting mechanism analyzing visual consistency, progress stages, duplicate images, and potential mismatches between reported progress and visual evidence.

10. 📊 Progress vs Expenditure Analysis

Compares physical progress directly against financial progress.

Physical progress: 40% | Expenditure: 85% → **High-Risk Financial Mismatch**

11. ⌚ Delay Detection & Prediction

Provides early warnings by detecting significantly delayed projects and predicting completion probabilities based on current trajectory.

12. 🔍 Duplicate & Overlapping Work Detection

Detects potentially duplicate projects using project IDs, location, NLP description similarity, cost, contractor, and time periods.

15. 🧠 AI Risk Scoring Engine

Combines multiple signals into a transparent, explainable project-level risk score (e.g., 84/100 — High Risk).

16. 🔮 Predictive Analytics

Forecasts potential future bottlenecks, cost overruns, and recurring agency patterns.

05 — Decision Support & Inspection Management

Pillar 5

14. 📋 Automated Compliance Monitoring

Checks availability and consistency of documents, sanction records, bills, milestones, and expenditure limits (Outputs:  Compliant,  Attention Required,  Verification Required).

17. 🚨 Intelligent Alert & Notification System

Generates severity-tiered alerts ( Normal,  Watch,  Review Required,  High Risk) routed exclusively to responsible authorities.

18. 💡 Explainable Recommendations

Provides transparent insights for anomalies: What happened → Why it's unusual → Supporting evidence → Recommended check.

21. 📝 Digital Inspection & Verification

Closes the loop between AI alerts, physical field inspections, and administrative corrective actions.

13. Contractor & Implementing Agency Performance Analytics

Builds performance intelligence based on historical project execution (completed on time, delays, cost/invoice anomalies) to guide administrative oversight.

19. Multi-Level Monitoring Dashboards

Tailored analytical dashboards across governance tiers:

- **MoSPI Dashboard:** India-wide overview, state/district rollups, fund utilization, and risk trends.
- **State Dashboard:** State-wide comparison, district performance, and delay tracking.
- **District Dashboard:** Operational management, active works, bills, evidence, and verification alerts.
- **MP Dashboard:** Constituency-level recommended works, approval status, expenditure, and anomalies.

22. Complete Audit Trail

Records every critical action (Who → Did what → When → Which project → What changed → Why) across recommendations, sanctions, evidence, and administrative decisions.